

Bachelor of Information Systems and Technology

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Programme Name and Duration

The programme name is Bachelor of Information Systems and Technology (BIST) and is hosted at the Department of Information Systems. The programme runs for 3 academic years comprising of six (6) semesters.

1.1 Target Group

The program targets two categories of entry schemes, namely Direct Entry and Diploma holders in relevant programs.

1.2 Nature of the Program

The day program has both government and privately sponsored students. The evening program has privately sponsored students only.

1.3 Duration

The duration of this program is three (3) academic years consisting of six semesters and two recess terms. Each semester lasts for seventeen (17) weeks out of which two

- weeks are for The first and second years will in addition have a recess term of 10 weeks. For students without the Cisco Certified Networking Associate (CCNA) certification will take CCNA as an audited course during the first year recess term.

1.4 Tuition Fees

Tuition fees for privately sponsored students shall be US\$ 3,200,000 Uganda Shillings per year for Ugandans (see appendix) and East African students and US\$ 4,700,000 Uganda Shillings per year for International Students.

Programme Description

The Bachelor of Information Systems and Technology (BIST) is a three year full-time Program open to all qualifying students, commencing in the 2017/2018 academic year.

The Program will be based on an effective curriculum, achieved through considering a number of factors that cover the opportunities, and gaps that need to be addressed by the BIST Program. These included;

1. The requirements from the inter university council for East Africa's March 2015 benchmarks for developing curricula for bachelor of Information Technology Programs
2. The internationally recognized ACM curriculum development guidelines for undergraduate degree programs in information systems and information technology
3. A market analysis to evaluate the attractiveness and the dynamics of the Information Systems and Technology market within East Africa. Special attention was paid on the kind of jobs/roles being advertised within this market segment and the different skills set that are being sought
4. Feedback obtained over a three year period (2013 – 2016) from our industrial partners where our students were placed for the Internship / field
5. Consultations made with the ICT Association of
6. Industrial and alumni feedback: key stakeholders and experts from regulatory authorities, industry and academia were invited to provide insights as to what their expectations are of an Information Systems and Technology.

The emphasis of this Program is on the ability of graduates to develop conceptual and practical IT skills and their application in communication, business problem solving and integration of organizational functions within the overall business strategy. The BIST program is thus designed to equip learners with the concepts, knowledge and skills required to build and manage robust computer information systems and electronic services that are properly aligned with organizational business strategy.

Graduates of this Program will take up positions related to functionalizing the Information and Communication Technology (ICT) domain in public and private sectors. Examples of these positions include: Web-Designers, Web-Programmers, Information Technology Planners, Systems Analysts, IT Project Managers, IT systems auditors, Information Systems Specialists, Security Specialists, Systems Administrators, Network Developers, Network Engineer and Forensics, Network Administration, Programming, Information Technology and Assurance. These competences cover the three broad areas of Information Technology Security, Systems Development and Information Systems Management.

Programme Rationale	Programme Objectives	Program Learning Outcomes	Entry Requirements
All aspects of the global computing field continue to face rapid and frequent change. The professional context in which the BIT and BIS graduate works has changed considerably over the decades and curricula need to reflect this change. University level curricula need frequent cyclic change to remain effective. The merger of the Bachelor of Information technology and the Bachelor of Information Systems programs at SCIT reflects the need for the following: • It is a requirement by the National Council for Higher Education (NCHE) for a periodic review of curricula based on program cycles: considering that the BIT and BIS programs were last reviewed in They had run for a cycle of three • years, and were due for review. The proposed BIST program is in line with this • A compelling argument for merging the two programs into one Bachelor of Information Systems and Technology (BIST) is a response to strong industry Feedback from internship evaluation reports of the students by industry stakeholders consistently revealed a need by the IT and IS job market for more system developers, system security experts and information systems managers. Furthermore, the students from the two programs were not a ready workforce as characteristically they had strong theoretical knowledge but less skilled on the practical applicability of this knowledge. The BIST program is therefore demand driven and tailored to bridge this gap as well. • There is an increased demand for graduates equipped with knowledge and skills critical for success in the 21st century such as; life and career skills; learning and innovation skills; and information, media and technology The BIST program (with its three areas of specializations) provides learners with essential skills to be at the cutting edge in the rapidly evolving digital economy. The curriculum is designed to educate graduates who are prepared to enter the workforce equipped with relevant knowledge, skills and competencies. The BIST program design separates the core of the curriculum from the electives with the intent of supporting the concept of career tracks. The program is therefore designed to have a set of selected course units that enable a graduate to apply IT and IT skills in solving societal and organizational problems in such diverse fields as medicine, agriculture, commerce, tourism and environment among many others. The mode of delivery used in the program provides for learning, life, and career skills such as critical thinking, communication, collaboration, and creativity.			

Programme Structure

Cores: (5 core courses)									
IST 1101	Foundations of Information Systems and Technology	45	30	—	60	4	Core	Modified	IS
CSK 1101	Communication Skills	45	30	—	60	4	Core	Old	Lang.
IST 1102	Emerging Trends in Information Systems and Technology	45	—	30	60	4	Core	Modified	IS
CSC 1107	Structured Programming	45	30	—	60	4	Core	Old	CS
MTH 1110	Basic Mathematics	30	—	30	45	3	Core	Old	MTH
Audited course:									
CSC 1109	Computer Literacy	30	60	—	60	4	Audited	Old	CS
Total CU						19			

Year 1 Semester II (5 Courses)

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 1201	Applied Business Statistics	30	30	30	60	4	Core	Modified	IS
IST 1202	Introduction to Computer Networks	45	30	—	60	4	Core	Old	IS
IST 1203	Data and Information Management I	30	60	—	60	4	Core	Modified	IS
IST 1204	Systems Analysis and Design	30	30	30	60	4	Core	Modified	IS
CSC 1214	Object Oriented Programming	30	60	—	60	4	Core	Old	CS
Total CU						20			

Year 1: Recess Term

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
CSC 1303	Cisco Certified Network Associate	—	150	—	75	5	Audited	Old	CS
CSC 1304	Practical Skills Development	15	90	30	75	5	Core	Old	CS
Total CU						5			

Year 2 Semester I (5 Courses)

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 2101	Data and Information Management II	30	60	—	60	4	Core	Old	IS
IST 2102	Web Systems and Technologies I	45	30	—	60	4	Core	Old	IT
BAM 2102	Entrepreneurship Principles	30	—	30	45	3	Core	Modified	Bus
IST 2103	Information Systems Security and Risk Management	30	30	—	45	3	Core	Old	IS
IST 2104	Electronic Media Systems and Multimedia	45	30	—	60	4	Core	Old	IT
Total CU						18			

Year 2 Semester II (5 Courses)

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 2201	System Administration	45	30	—	60	4	Core	Modified	IT
IST 2202	E-services	30	30	—	60	4	Core	Modified	IS
IST 2203	Research Methodology	45	—	30	60	4	Core	Old	IT
IST 2204	IST Project Management	45	30	—	60	4	Core	Old	IT
IST 2205	Web Systems and Technologies II	30	30	30	60	4	Core	Old	IT
Total CU						20			

Year 2: Recess Term

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
IST 2302	Field Attachment	—	150	—	75	5	Recess	Old	IS
Total CU						5			

Year 3 Semester I (5 Courses) – Information Technology Security Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 3101	Project I	45	30	30	75	5	Core	Modified	IT
IST 3102	Network Security	30	30	30	60	4	Core	New	IT
IST 3103	Enterprise Network Management	45	30	—	60	4	Core	Modified	IT
IST 3104	Intrusion Detection and Incident Management	45	30	30	60	4	Core	New	IT
IST 3105	Digital Forensics Investigation	45	30	30	60	4	Core	Modified	IT
Total CU						21			

Year 3 Semester II (4 Courses) – Information Technology Security Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (3 core courses)									
IST 3201	Project II	—	100	50	75	5	Core	Modified	IT
IST 3202	Ethical Hacking	45	30	—	60	4	Core	New	IT
IST 3203	IT Law and Ethics	30	—	30	45	3	Core	Old	IT
Electives: (1 Course)									
IST 3204	Information Systems Audit	30	30	—	45	3	Core	Old	IS
IST 3205	Software Systems Testing	30	30	—	45	3	Core	Old	IS
Total CU						15			

Year 3 Semester I (5 Courses) – Systems Development Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 3101	Project I	45	30	30	75	5	Core	Modified	IT
IST 3106	Integrative Programming and Technologies	30	30	30	60	4	Core	Old	IT
IST 3107	Intelligent Systems	45	30	—	60	4	Core	Modified	IS
IST 3108	Applications Development	30	30	30	60	4	Core	Modified	IS
CSC 3110	User Interface Design	30	30	30	60	4	Core	Modified	CS
Total CU						21			

Year 3 Semester II (4 Courses) – Systems Development Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (3 core courses)									
IST 3201	Project II	—	100	50	75	5	Core	Modified	IT
IST 3203	IT Law and Ethics	30	—	30	45	3	Core	Old	IT
IST 3206	System Integration and Deployment	30	30	30	60	4	Core	Old	IT
Electives: (1 course)									
IST 3204	Information Systems Audit	30	30	—	45	3	Core	Old	IS
IST 3205	Software Systems Testing	30	30	—	45	3	Core	Old	IS
Total CU						15			

Year 3 Semester I (5 Courses) – Information Systems Management Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (5 core courses)									
IST 3101	Project I	45	30	30	75	5	Core	Modified	IT
IST 3108	Applications Development	45	30	—	60	4	Core	Modified	IS
IST 3109	Data Warehousing and Business Intelligence	30	30	30	60	4	Core	Old	IS
IST 3110	Business Process Management	30	30	30	60	4	Core	Old	IS
IST 3111	Information Systems Architecture	45	30	—	60	4	Core	Modified	IS
Total CU						21			

Year 3 Semester II (4 Courses) – Information Systems Management Option

Course Code	Course Name	LH	PH	TH	CH	CU	Type	Remark	Host dept
Cores: (3 core courses)									
IST 3201	Project II	—	100	50	75	5	Core	Modified	IT
IST 3203	IT Law and Ethics	30	—	30	45	3	Core	Old	IT
IST 3207	Information Systems Strategy, Management and Acquisition	45	30	—	60	4	Core	Old	IS
Electives: (1 Course)									
IST 3204	Information Systems Audit	30	30	—	45	3	Core	Old	IS
IST 3208	Modelling and Simulation	30	30	—	45	3	Core	Old	IS
Total CU						15			

The Graduation Load

The minimum graduation load is one hundred twenty three (123) credit units. The core courses in the first four semesters (year 1 plus year 2) contribute eighty seven (87) credit units to the graduation load. Each of the three tracks (in year 3) contributes thirty six (36) credit units to the graduation load.

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